

Driver Drowsiness & Yawn Detection System

AI / Computer Vision / Deep Learning Project

Developed

by

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Project Overview

This project is an AI-based Driver Drowsiness and Yawn Detection System. It monitors a driver's facial behavior through a webcam and identifies two key signs of fatigue: prolonged eye closure and yawning. A custom Convolutional Neural Network (CNN) was trained from scratch on a labeled dataset of eye and mouth states, then deployed through a live, interactive web interface where predictions and confidence scores are shown in real time.

Main Objectives

- Detect eye closure and yawning from the driver's facial features.
- Monitor the driver continuously through a live webcam feed.
- Flag possible drowsiness with a clear, real-time alert and confidence score.
- Provide an accessible, browser-based interface with no local setup required to try it.
- Demonstrate a complete applied deep learning and computer vision pipeline, from raw data to a working demo.

Technologies Used

- Python
- TensorFlow / Keras (CNN with Batch Normalization for stable training)
- OpenCV (face, eye, and mouth region detection via Haar Cascades)
- Gradio (real-time web interface)
- Google Colab (training and deployment environment)
- Webcam for live real-time detection

How the System Works

1. The webcam captures a live video frame of the driver's face.
2. A face detector locates the face, then separately isolates the eye region and the mouth region.
3. Each region is passed through the trained CNN, which classifies eye state (Open/Closed) and mouth state (Yawn/No Yawn) independently.
4. If the eyes are detected closed, or a yawn is detected, the system raises a DROWSY alert along with the model's confidence percentage.
5. Otherwise, the system reports the driver as ALERT.

Model Development Process

The model was trained using an augmented dataset (rotation, zoom, shift, brightness variation) to reduce overfitting, combined with Batch Normalization and a reduced learning rate for training stability. Two earlier training attempts failed — one from severe overfitting (97% training accuracy vs. 65% validation accuracy) and one from full model collapse (predicting a single class regardless of input, 25% accuracy). Both issues were diagnosed and resolved through architecture and training adjustments, resulting in a stable final model.

Model Performance

The final CNN achieved an overall validation accuracy of approximately 73-80% across four classes (Open, Closed, Yawn, No Yawn).

- Eye-state detection (Open/Closed): high precision and recall, reliable in real-time testing.
- Yawn detection (Yawn/No Yawn): functional but less consistent — a known limitation, noted as an area for future improvement (e.g. more training data, a dedicated mouth-detection model).

Live Web Interface

The trained model is deployed through a Gradio-based web application, allowing anyone to test the system directly in a browser using their own webcam. The interface displays:

- A live camera feed
- Real-time Eye State and Mouth State predictions with confidence percentages
- A color-coded status card (green = Driver Alert, red = Drowsy)
- A Capture Snapshot button to freeze and save the current frame for demo purposes

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Expected Outcome

The final system provides a real-time, browser-accessible demonstration of driver fatigue monitoring, combining two independent detection signals into a single alert system. This project demonstrates the practical application of deep learning, computer vision, and interactive deployment in the context of road-safety technology.



